

Summer 2026 Grassland Soil Core Figures

Created: August 16, 2026 Last Updated:

Jared Umphress

Table of contents

Part 1: Set up tasks	1
Part 2: Convert EC measurements to dS/m	2
2.1: Get mean EC and soil moisture values for depth midpoints	2
Part 3: Plot EC and soil moisture based on transect	3

Part 1: Set up tasks

Load in libraries

```
library(tidyverse)
library(janitor)
library(here)

#plotting
library(cowplot)
library(ggthemes) # to use Paul Tol color palettes

# packages specifically for soil science
library(aqp) # algorithms for quantitative pedology
# library(soiltexture)
```

Load in data

```
soil_data <- read.csv(here("data/soil_samples/2026_summer_grassland_soil_cores.csv"))
```

Part 2: Convert EC measurements to dS/m

This function also accounts for dilution

```
source(here("code/mS_per_cm_to_dS_per_m.R"))

#clean data, get depth midpoints, and estimate EC in dS/m
soil_data <- soil_data %>%
  clean_names() %>%
  mutate(
    depth_midpoint = -(depth_top + depth_bottom)/2,
    ec_ds_per_m = uS_per_cm_to_dS_per_m(ec_u_s_cm),
    ec_converted = (ec_u_s_cm/1000)*5,
    year = year(sample_date)
  )
```

Test uS/cm to dS/cm conversion

```
# high value of 2440
uS_per_cm_to_dS_per_m(2440)
```

```
[1] 16.451
```

```
# 16.451 dS/cm
```

```
# low value of 235 uS/cm
uS_per_cm_to_dS_per_m(235)
```

```
[1] 3.0005
```

```
# 3.005 dS/cm
```

2.1: Get mean EC and soil moisture values for depth midpoints

```
#calculate mean values for both plots combined
means_2026_summer <- soil_data %>%
  group_by(depth_midpoint) %>%
  summarize(mean_ec = mean(ec_ds_per_m, na.rm = TRUE),
             mean_moisture = mean(percent_soil_moisture),
             #mean_sand = mean(sand),
             #mean_clay = mean(clay),
             n = n()) %>%
  ungroup()
```

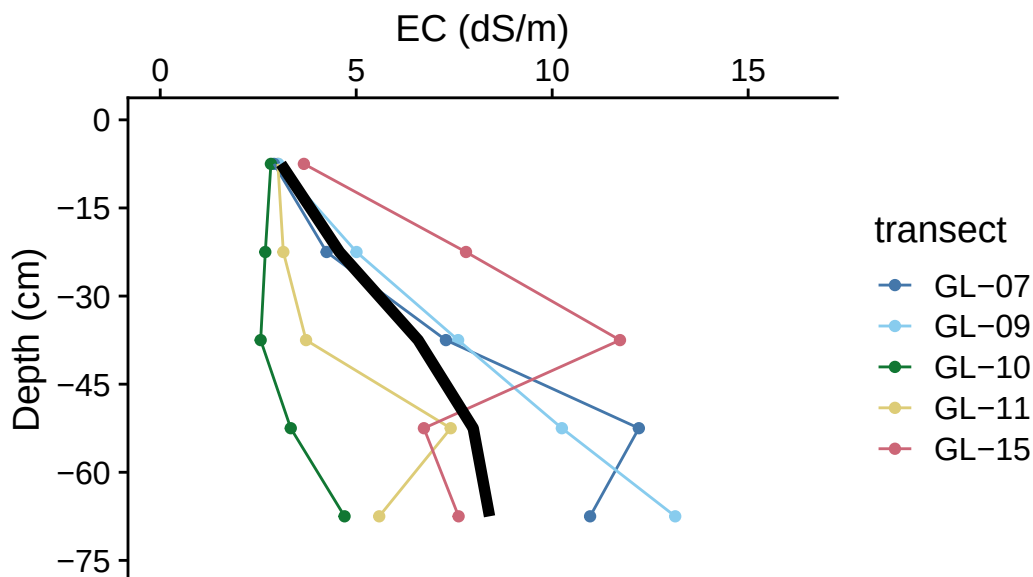
Part 3: Plot EC and soil moisture based on transect

EC figures

```
fig_2026_summer_ec <- ggplot(data = soil_data, aes(x = depth_midpoint, y = ec_ds_per_m, color = percent_soil_moisture)) +
  geom_point() +
  geom_path() +
  geom_path(data = means_2026_summer, aes(x = depth_midpoint, y = mean_ec), color = "black", size = 2) +
  coord_flip() +
  theme_cowplot() +
  scale_x_continuous(limits = c(-75,0), breaks = seq(-75, 0, by =15)) +
  scale_y_continuous(limits = c(0,NA), position = "right") +
  labs(y = "EC (dS/m)", x = "Depth (cm)", title = "Mesa Grassland Cores, July 2026") +
  scale_color_ptol()

fig_2026_summer_ec
```

Mesa Grassland Cores, July 2026

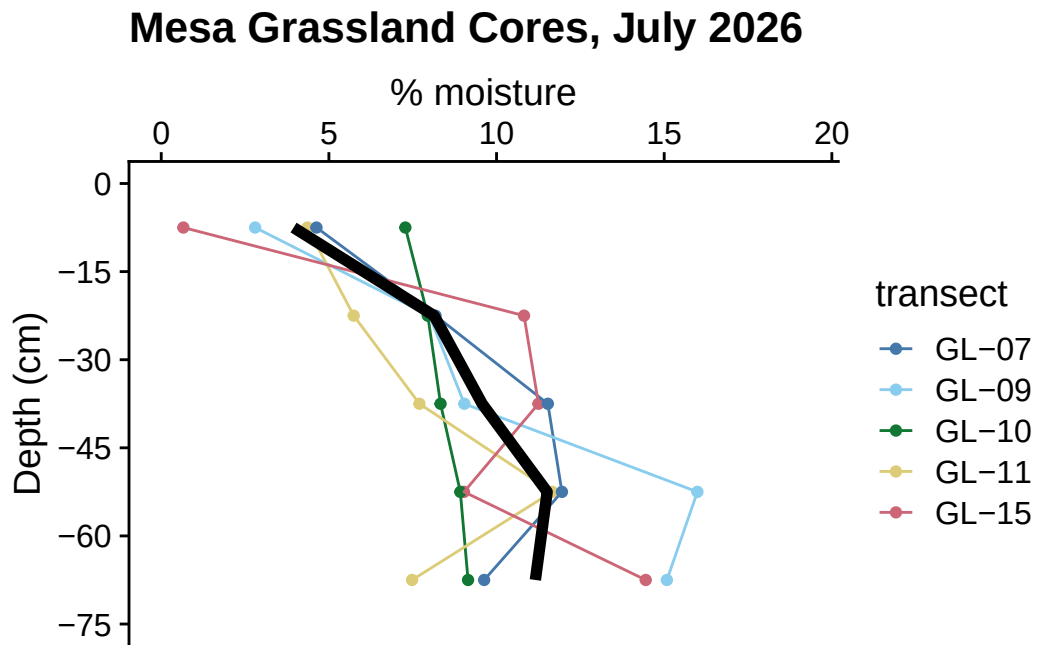


```
#save to file
ggsave(filename = here("figures/soil_cores/Summer_2026_Mesa_Grassland_EC.pdf"),
        fig_2026_summer_ec,
        width = 6,
        height = 4,
        units = "in",
        bg = "white")
```

Soil Moisture figures

```
# moisture plot ----
fig_2026_summer_moisture <- ggplot(data = soil_data, aes(x = depth_midpoint, y = percent_soil_moisture)) +
  geom_point() +
  geom_path() +
  # plot thick black line for mean values
  geom_path(data = means_2026_summer, aes(x = depth_midpoint, y = mean_moisture), color = "black", size = 2) +
  coord_flip() +
  theme_cowplot() +
  scale_x_continuous(limits = c(-75,0), breaks = seq(-75, 0, by = 15)) +
  scale_y_continuous(limits = c(0,NA), position = "right") +
  labs(y = "% moisture", x = "Depth (cm)", title = "Mesa Grassland Cores, July 2026") +
  scale_color_ptol()
```

fig_2026_summer_moisture



```
ggsave(filename = here("figures/soil_cores/Summer_2026_Mesa_Grassland_moisture.pdf"),  
  fig_2026_summer_moisture,  
  width = 6,  
  height = 4,  
  units = "in",  
  bg = "white")
```